Computational Tools Inventory for AI-Based Scientific Replication

A Survey of 31 DOE-Funded Research Papers

REPLICATE Project

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Abstract

This document catalogs all computational tools, software packages, libraries, databases, and frameworks required to replicate 31 scientific papers identified as candidates for AI-based replication from the OSTI corpus of ∼67,000 DOE-funded publications. Papers 1–11 were identified in an initial broad screen; papers 12–31 were selected in a focused search for **high AI-replicability** candidates that use only open-source tools, are purely computational, and have clear quantitative outcomes.

Contents

1 Paper Inventory

#	OSTI ID	Title (abbreviated)	Domain	OSS	Repo
Initial screen (Papers 1–11)					
1	1997354	Integer Sequences from Hausdorff Metric Geometry	Combinatorics		
2	2571540	Portfolio Approach to Parallel Bayesian Optimization	Optimization / ML		
3	1609039	Deformation of $Cu_{64}Zr_{36}$ via MD	Molecular Dynamics		
4	1981773	Pt Doping on $La_2Ti_2O_7$: DFT Study	DFT		
5	2217719	SCALE Depletion for Molten Salt Reactors	Nuclear Eng.		
6	1475143	FDTD: Solving 1+1D Delay PDE in Parallel	Comp. Physics		
7	1983793	Coplanar Waveguide Resonator FEM	EM Simulation		
8	1606674	CMV Reduction in Quasi-Z-Source Inverter	Power Electronics		
9	2475938	Updated Virophage Taxonomy	Bioinformatics		
10	1427646	Deep Learning of STEM Images	Deep Learning		
11	—	AI-Driven Passivation Molecules for PSCs	Generative AI / DFT		
High-replicability screen (Papers 12–31)					
12	1578031	fldgen v2.0: Earth System Model Emulation	Climate / Statistics		
13	1461824	Approximating Photo-z PDFs (qp)	Astrophysics / ML		
14	2582579	NILC Power Spectrum for Cosmological Parameters	Cosmology		
15	1624105	Linclust: Clustering Protein Sequences	Bioinformatics		
16	3014512	Spin-Dependent Dark Matter Scattering (DarkELF)	Particle Physics		
17	2439897	Invariant Variational Autoencoders for Microscopy	ML / Materials		
18	2587225	ScaWL: Scaling k-WL Graph Algorithms	Graph Algorithms		
19	1861801	NukeLM: Nuclear Domain Language Models	NLP / ML		

#	OSTI ID	Title (abbreviated)	Domain	OSS	Repo
20	1984484	DRAS: Deep RL for HPC Scheduling	Reinforcement Learning		
21	2469515	Genome Extraction from Metagenomics (PATRIC)	Bioinformatics		
22	1523841	Shift Photocurrent from Polarization Differences	Condensed Matter Theory		
23	1864334	VMC of A$\leq$4 Nuclei with Neural Networks	Nuclear Physics / ML		
24	2441075	Light-Scattering Errors in Trapped-Ion Qubits	Quantum Simulation		
25	1379592	Mathematical Foundations of Graph-BLAS	Graph Theory / LA		
26	1565592	Markov State Models from Non-Equilibrium Sims	Comp. Chemistry		
27	2396968	Latent SDEs for Quasar Variability	Astrophysics / ML		
28	1412756	Chiral Spin Order in Kondo-Heisenberg Systems	Condensed Matter Theory		
29	1842593	Motion Tomography via Occupation Kernels	Applied Math		
30	1993311	Electronic Heat Capacities via DMQMC + GPR	Quantum Chemistry		
31	2571909	Hybrid ML for Critical Heat Flux Prediction	Thermal-Hydraulics / ML		

Table 1: All 31 papers selected for AI-based replication. OSS = all tools open-source; Repo = code repository available.

2 Complete Tool Inventory by Category

2.1 Simulation and Modeling Codes

Tool	Paper(s)	Description
`LAMMPS`	#3	Molecular dynamics simulation
`VASP`	#4, #11	Vienna Ab Initio Simulation Package (DFT)
`SCALE 6.3`	#5	ORNL nuclear systems code suite
`Serpent`	#5	Monte Carlo neutron transport
`MATLAB/Simulink`	#8	Circuit and system simulation
`HFSS/COMSOL`	#7	Finite-element EM solver
`Mathematica`	#6	Symbolic math
`ACEMD`	#26	MD simulation (alanine dipeptide)
`Sim5`	#27	GR ray-traced accretion disk simulator
`HANDE-QMC`	#30	Open-source quantum Monte Carlo
`Molpro`	#30	Quantum chemistry (HF integrals)
`CQSim`	#20	Event-driven HPC job scheduler simulator
`metaSPAdes`	#21	Metagenome assembler

Table 2: Simulation and modeling codes.

2.2 Programming Languages

Language	Paper(s)
`Python`	#5, #6, #9, #10, #11, #13, #14, #16, #17, #19, #20, #23, #27, #30, #31
`R`	#2, #9, #12
`C/C++`	#6, #8, #15, #18, #25
`Fortran`	#24 (Numerical Recipes reference)
`CUDA`	#25 (GraphBLAS GPU benchmarks)

Table 3: Programming languages across all 31 papers.

2.3 Machine Learning and Deep Learning Frameworks

Tool	Paper(s)	Description
`PyTorch`	#17, #20, #23, #27	Deep learning framework
`TensorFlow`	#10, #20, #31	Deep learning framework
`TensorFlow Probability`	#31	Bayesian neural networks, deep GPs
`Keras`	#10	High-level DL API
`GPT-2` (Transformers)	#11	Generative language model
`LLaMA-2`	#11	Alternative generative model
`SMILES-X`	#11	SMILES-based molecular classifier
`HuggingFace Transformers`	#19	RoBERTa/SciBERT pre-training

BERTopic	#19	Topic modeling on embeddings
HDBSCAN	#19	Density-based clustering
scikit-learn	#11, #13	Classical ML algorithms
scikit-image	#10, #17	Image processing
Ray Tune	#31	Hyperparameter optimization (ASHA)
AtomAI	#17	VAE for microscopy images
pyroVED	#17	VAE construction toolkit
GPy	#30	Gaussian process framework
GPyTorch	#27	Gaussian process in PyTorch
BoTorch	#27	Bayesian optimization
torchsde	#27	Neural SDE solver
Celerite	#27	Fast GPR for time series
RDKit	#11	Cheminformatics toolkit
UMAP	#11, #19	Nonlinear dimensionality reduction
NetworkX	#10	Graph analysis

Table 4: Machine learning, deep learning, and AI tools.

2.4 Parallel Computing Libraries

Library	**Paper(s)**	**Description**
OpenMP	#6, #15, #18	Shared-memory parallelism
MPI	#6, #18	Distributed-memory parallelism
POSIX Threads	#6	Threading
CUDA	#25	GPU computing

Table 5: Parallel computing libraries.

2.5 R Packages

Package	**Paper(s)**	**Purpose**
fldgen	#12	Earth system model emulation
ncdf4	#12	NetCDF file I/O
hetGP	#2	Heteroscedastic GP modeling
DiceOptim	#2	Batch Expected Improvement
GPareto	#2	Multi-objective batch EHI
DiceKriging	#2	Deterministic GP modeling
mlrMBO	#2	Model-based optimization
mco	#2	NSGA-II
pso	#2	Particle swarm optimization
ggplot2	#2, #9, #12	Visualization
dplyr	#9, #12	Data manipulation
tidyr	#9, #12	Data tidying
ggtree	#9	Phylogenetic tree visualization
reshape2	#12	Data reshaping

Table 6: R packages.

2.6 Python Scientific Libraries

Library	Paper(s)	Purpose
NumPy	#13, #14, #27, #30	Numerical computing
SciPy	#13, #14, #27, #30	Scientific computing
matplotlib	#13, #14, #27	Plotting
Astropy	#14, #27	Astronomy utilities
pandas	#30	Data manipulation
SymPy	#30	Symbolic mathematics
pathos	#13	Parallelization
corner.py	#27	Corner plots
astroML	#27	Astronomical ML utilities
pywigxjpf	#14	Wigner 3j symbols
HEALPix/healpy	#14	Spherical harmonic operations

Table 7: Python scientific libraries.

2.7 Bioinformatics Tools

Tool	Paper(s)	Purpose
MMseqs2/Linclust	#15	Linear-time protein clustering
hmmsearch	#9	HMM profile searching
BLAST+	#9, #15, #21	Sequence comparison
MAFFT	#9	Multiple sequence alignment
MUSCLE	#9	Multiple sequence alignment
DIAMOND	#9, #15	Fast protein alignment
IQ-Tree	#9	Phylogenetic tree building
clipkit	#9	Alignment trimming
Prodigal	#9, #15	Gene prediction
hhblits/hhsearch	#9	HMM–HMM comparison
MCL	#9	Markov clustering
InfoMap	#9	Network clustering
MUMmer	#9	ANI-based dereplication
CheckV	#9	Viral genome QC
EasyFig	#9	Genome map visualization
CD-HIT	#15	Protein clustering (benchmark)
MetaBAT2	#21	Unsupervised metagenome binning
Bowtie2	#21	Read alignment
RASTtk	#21	Genome annotation
CheckM	#21	Genome completeness assessment
PyEMMA	#26	Markov state model analysis

Table 8: Bioinformatics and computational chemistry tools.

2.8 Databases and Data Sources

Database	Paper(s)	Description
OEIS	#1	Online Encyclopedia of Integer Sequences
NCBI nr/GenBank/RefSeq	#9, #21	Nucleotide and protein databases
IMG/VR v3	#9	Virus genome database
Pfam	#9, #15	Protein family database
UniProt	#15	Universal protein database
Metaclust	#15	424M protein sequence clusters
NCBI SRA	#15, #21	Sequence Read Archive
ENDF/B-VII.1	#5	Nuclear data
IRPhEP	#5	Reactor physics benchmarks
MNIST	#2, #17	Handwritten digit dataset
PubChem	#11	Chemical compound database
OSTI	#19	DOE technical reports (∼1.5M abstracts)
PATRIC	#21	Pathosystems genome database
NRC CHF Database	#31	Critical heat flux lookup table
WebSky CMB Mocks	#14	Extragalactic CMB simulations
SuiteSparse Matrix Coll.	#18	Sparse matrix benchmarks
Buzzard mock catalog	#13	Galaxy mock catalog
SDSS / LSST	#13, #27	Sky survey data
Gliderpalooza 2013	#29	Ocean glider trajectory data
HPC workload traces	#20	Theta (ALCF), Cori (NERSC) job logs

Table 9: Databases and data sources.

2.9 Open-Source Code Repositories

Repository URL	Paper
https://github.com/leofang/FDTD	#6
https://github.com/Boulder-Cryogenic-Quantum-Testbed/simple-resonator-mask	#7
https://github.com/simroux/ICTV_VirophageSG	#9
https://github.com/adroitfajar/pvmol-gen	#11
https://github.com/jgcri/fldgen	#12
https://github.com/aimalz/qp	#13
https://github.com/kmsurrao/NILC-PS-Model	#14
https://github.com/soedinglab/mmseqs2	#15
DarkELF (github.com/tongylin/DarkELF)	#16
https://github.com/saimani5/VAE-tutorials	#17
https://github.com/CobySoss/ScaWL	#18
https://github.com/SPEAR-IIT/CQGym	#20
https://github.com/SEEDtk/p3_code	#21
https://code.ornl.gov/5ci/exactly-solved-model...	#24

http://pyemma.org	#26
https://github.com/JFagin/latent_SDE	#27

Table 10: Open-source code repositories (16 of 31 papers).

3 Per-Paper Tool Summary

#	Paper	Key Tools
1	Integer Sequences	CAS (Mathematica/SageMath), OEIS
2	Bayesian Opt. Portfolio	R, hetGP, DiceOptim, GPareto, mlrMBO, mco, pso
3	$Cu_{64}Zr_{36}$ MD	LAMMPS, Cheng EAM potential, Voronoi, OVITO
4	$Pt/La_2Ti_2O_7$ DFT	VASP, PBE/PBE0, DFT-D3, Bader, VESTA
5	SCALE MSR	SCALE 6.3, Serpent, ChemTriton (Python), ENDF/B-VII.1
6	FDTD Delay PDE	C99, OpenMP, pthreads, GCC, Mathematica
7	CPW Resonator	HFSS/COMSOL, GDS mask (GitHub)
8	CMV Inverter	MATLAB/Simulink
9	Virophage Taxonomy	∼20 bioinformatics tools, R, NCBI
10	STEM Deep Learning	Keras/TensorFlow, scikit-image, NetworkX
11	Perovskite AI	GPT-2, SMILES-X, RDKit, VASP, PubChem
12	fldgen v2.0	R (≥3.3.3), fldgen, ncdf4, ggplot2, dplyr, tidyr
13	Photo-z PDFs (qp)	Python, qp, NumPy, SciPy, scikit-learn, matplotlib
14	NILC Cosmology	Python, HEALPix/healpy, pywigxjpf, NumPy, SciPy, astropy
15	Linclust	C/C++, MMseqs2, OpenMP, SSW library, UniProt
16	DarkELF DM	Python, DarkELF
17	Invariant VAEs	Python, PyTorch, AtomAI, pyroVED, MNIST
18	ScaWL	C/C++, MPI, OpenMP, GCC
19	NukeLM	Python, HuggingFace, RoBERTa, BERTopic, OSTI data
20	DRAS RL Scheduling	Python, TensorFlow, PyTorch, OpenAI Gym, CQ-Gym/CQSim
21	PATRIC Metagenomics	metaSPAdes, MetaBAT2, Bowtie2, RASTtk, BLAST, CheckM
22	Shift Photocurrent	None (pure analytic; any language suffices)
23	VMC Neural Networks	Python, PyTorch (or JAX), VMC algorithm
24	Trapped-Ion Errors	Fortran/Python, Lindbladian solver
25	GraphBLAS Foundations	C/C++, CUDA, GraphBLAS, GCC, nvcc
26	Markov State Models	Python, PyEMMA, ACEMD, AMBER force field
27	Latent SDEs (Quasars)	Python, PyTorch, torchsde, BoTorch, GPyTorch, Sim5
28	Chiral Spin Order	None (pure analytic theory)
29	Motion Tomography	Python/MATLAB (RK4, Simpson's rule, RBF kernels)
30	DMQMC + GPR	Python, HANDE-QMC, GPy, NumPy, SciPy, pandas, SymPy

#	Paper	Key Tools
31	Hybrid ML for CHF	Python, TensorFlow, TF Probability, Ray Tune, NRC CHF data

Table 11: Per-paper tool summary for all 31 papers.

4 Tool Statistics

4.1 Language Prevalence

- **Python** — 15 of 31 papers (48%)
- **R** — 3 papers (10%)
- **C/C++** — 5 papers (16%)
- **Pure analytic** (any language) — 3 papers (10%)

4.2 AI Replicability Tiers

Tier	Papers	Criteria
HIGH (fully automatable)	20	Open-source only, code repos, public/synthetic data
MEDIUM (partial automation)	6	Open-source but needs HPC, domain expertise, or data curation
LOW (manual steps required)	5	Licensed software (VASP, SCALE, MATLAB, COMSOL)

5 Installation Complexity

5.1 Tier 1: Straightforward (pip / conda / CRAN)

- Python / R / Julia
- PyTorch / TensorFlow
- Keras
- HuggingFace Transformers
- scikit-learn / scikit-image
- NetworkX
- GPT-2 / SMILES-X
- RDKit
- UMAP / BERTopic
- NumPy / SciPy / pandas
- matplotlib / Astropy
- SymPy / GPy / GPyTorch
- BoTorch / torchsde
- Ray Tune
- TF Probability
- AtomAI / pyroVED
- fldgen / ncdf4
- HEALPix / healpy
- PyEMMA
- OpenAI Gym
- corner.py / astroML
- pathos / pywigxjpf
- All R packages (hetGP, DiceOptim, GPareto, etc.)
- All bioinformatics tools (BLAST+, HMMER, MAFFT, etc.)

5.2 Tier 2: Moderate

- `LAMMPS` — compile from source with EAM support
- `GCC/OpenMP/MPI` — compiler toolchain
- `MMseqs2/Linclust` — compiled C++ with SIMD
- `LLaMA-2` (7B) — GPU with $\geq$16 GB VRAM
- `HANDE-QMC` — Fortran/C compiled QMC code
- `GraphBLAS/GBTL` — C/CUDA compilation
- `Mathematica` — widely available academically

- `OVITO` — free basic version
- `Sim5` — GR ray-tracing (compile from source)

5.3 Tier 3: Licensed / Restricted

- `VASP` — academic license, Univ. Vienna (#4, #11)
- `SCALE 6.3` — RSICC distribution, export-controlled (#5)
- `Serpent` — academic license, VTT Finland (#5)
- `Ansys HFSS / COMSOL` — commercial FEM (#7)
- `MATLAB/Simulink` — MathWorks license (#8)
- `Molpro` — academic license (#30)

6 Recommended Environment Setup

1. **Python environment:** `conda create -n replicate python=3.10`
2. **Core ML:** `pip install torch tensorflow keras transformers scikit-learn scikit-image`
3. **Generative AI:** `pip install rdkit-pypi smilesx umap-learn bertopic`
4. **Scientific:** `pip install numpy scipy matplotlib astropy sympy pandas gpy gpytorch botorch torchsde`
5. **Domain:** `pip install pyemma healpy corner ray[tune] tensorflow-probability`
6. **R packages:** `install.packages(c("fldgen","ncdf4","hetGP","DiceOptim","GPareto","mlrMBO","ggp`
7. **Bioinformatics:** `conda install -c bioconda hmmer blast mafft diamond mmseqs2 iqtree prodigal checkm-genome metabat2 bowtie2`
8. **Compiled codes:** Build LAMMPS, FDTD, ScaWL, GraphBLAS from source
9. **Licensed codes:** Obtain VASP, SCALE, HFSS/COMSOL, MATLAB as needed
10. **GPU:** NVIDIA GPU with ≥16 GB VRAM for PyTorch/TensorFlow training